

AKILIU TAMIRAT ERKALO

AI Engineer & Full Stack Developer | JavaScript/TypeScript · LLMs · Agentic Systems

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PROFESSIONAL SUMMARY

Results-driven AI Engineer and Full Stack Developer with 4+ years of experience building scalable, production-grade web applications and actively expanding into AI systems engineering. Proficient in JavaScript/TypeScript, React, Next.js, Node.js, FastAPI, and modern LLM-powered architectures including LangChain, LangGraph, RAG pipelines, and autonomous agents. Experienced in evaluating AI-generated code outputs, prompt engineering, and integrating LLMs into production workflows. Strong grasp of JavaScript internals — async/await, closures, event loop, prototype chains — and AI evaluation concepts including reasoning validation, hallucination detection, and RLHF workflows. Proven track record of shipping products used by 100,000+ users in fintech, EdTech, and e-commerce.

TECHNICAL SKILLS

AI & LLMs	LangChain, LangGraph, RAG Pipelines, Prompt Engineering, LLM Evaluation, Agentic Workflows, RLHF concepts, Hallucination Detection, OpenAI API, Anthropic API
AI/ML Engineering	FastAPI, TensorFlow, Scikit-learn, Python, Pandas, Spark, Vector DBs, Embedding Models, Model Fine-tuning
JavaScript & TS	Advanced JavaScript (ES6+), TypeScript, Closures, Event Loop, Async/Await, Prototypes, Generics, Utility Types, Functional & OOP Patterns
Frontend	Next.js, React, TanStack, Astro.js, Tailwind CSS, Shadcn UI, SSR/CSR, Hydration, Responsive Design, Performance Optimization
Backend & APIs	Node.js, Express, NestJS, FastAPI, REST, GraphQL, Spring Boot, Secure API Design, Schema-Driven Architecture
Databases	PostgreSQL, MongoDB, MySQL, Prisma, Supabase, Neon, Vector DB
State Management	Zustand, Redux
DevOps & Tools	Git, Docker, Jenkins, ArgoCD, AWS, Vercel, Render, Webpack, Vite, Jest, CI/CD Pipelines

PROFESSIONAL EXPERIENCE

Frontend Engineer · [Safaricom](#) Apr 2021 – Present

Africa's leading telecom & fintech platform — building internal tools and M-Pesa customer-facing products at scale.

- ▶ Rebuilt the Merchant Onboarding Portal, enabling real-time coordination for 100+ nationwide field agents and measurably improving merchant management throughput.
- ▶ Engineered an E-commerce Management Dashboard that automated inventory control end-to-end, achieving a 99% boost in operational efficiency.
- ▶ Developed a Complaint Management Portal for M-Pesa customer support, reducing average ticket resolution time by 30%.
- ▶ Enhanced the M-Pesa Internal Support Portal, streamlining cross-departmental workflows for all M-Pesa operations.
- ▶ Reviewed and validated AI-generated code suggestions during development workflows, ensuring production-quality standards were maintained before deployment.
- ▶ Integrated LLM-based coding assistants critically — validating outputs, identifying hallucinations, and refining generated solutions for correctness and performance.

► **Project: Saving Mini-App — M-Pesa Super-App**

- Built and shipped a savings mini-application within the M-Pesa super-app, serving 100,000+ active users.
- Designed an intuitive UI/UX flow that improved financial engagement, built with Next.js, Zustand, and Tailwind CSS.
- Contributed to CI/CD pipelines (Jenkins + ArgoCD), ensuring reliable zero-downtime deployments.

Full Stack Developer · **SRE 22** May 2023 – Present

Software agency delivering custom platforms across higher education, retail, media, and affiliate marketing sectors.

- Optimized a Next.js application's performance by 25% in load time through code-splitting, lazy loading, and SSR tuning.
- Architected and co-built a University Management System automating 95% of administrative processes for students, faculty, and staff.
- Built a reusable component library in React + TypeScript, accelerating team development velocity by 30% and enforcing UI consistency.
- Integrated headless CMS platforms (Sanity, Strapi) reducing non-technical content update turnaround by 40%.
- Evaluated frontend and backend architecture decisions for correctness, maintainability, and scalability across multiple production applications.
- Conducted debugging and performance analysis for asynchronous JavaScript workflows, API integrations, and state management systems.

► **Project: Affiliate Program Management Platform (Esperanza)**

- Designed and built a centralized affiliate management system to onboard, track, and manage affiliates at scale with real-time dashboards.
- Implemented referral analytics, commission visibility, and reporting infrastructure designed for future scalability.

► **Project: Hooraflix Movie Streaming Platform**

- A multimedia streaming and digital engagement platform that delivers movies, live events, educational courses, and premium content through a single unified ecosystem. It also includes a hybrid affiliate system where users earn coins through referrals and unlock rewards, premium access, and growth opportunities.
- It leverages caching strategies and optimized backend architecture hosted on a dedicated VPS to ensure fast load times, low latency, and reliable streaming performance

► **Project: University Management System**

- Co-developed a full-featured platform automating 95% of administrative workflows across student, faculty, and staff roles.
- Implemented SSR for SEO, CMS integration, and Jest test coverage to ensure production reliability.

► **Project: E-Learning Web Application**

- Built a responsive platform with course browsing, enrollment, progress tracking, quiz modules, video playback, and payment integration.
- Integrated a CMS enabling teachers to independently upload and manage course materials.

► **Project: Video Streaming Platform**

- Developed a responsive video streaming web app with content browsing, playback optimization, and user interaction APIs.
- Focused on cross-device performance and maintainable component architecture.

► **Project: U-Shop E-Commerce Platform (Web, Mobile & Super App Integration)**

- Built a scalable e-commerce ecosystem featuring product browsing, cart management, order processing, and secure Mpesa payment integration, with seamless integration into the M-Pesa Super App. The platform includes a Flutter mobile application and a Java-based backend supporting high availability and transactional reliability.
- Developed the Backoffice Admin Portal using Next.js, enabling efficient management of products, orders, users, inventory, and analytics for operational teams. Collaborated across backend and mobile teams, focusing on system design, admin workflows, and ensuring smooth platform operations and business process control

Machine Learning Engineer (Intern) · **iCog-Labs** Mar 2025 – May 2025

Leading Ethiopian AI research lab — working on ML model evaluation, fraud detection, and predictive systems.

- Evaluated machine learning model outputs and improved fraud detection model accuracy by 10% via feature engineering and TensorFlow-based tuning.

- ▶ Built a Python/Spark data pipeline that reduced processing time by 50%, accelerating ML iteration cycles.
- ▶ Processed and analyzed large-scale datasets using Python, SQL, Pandas, and Spark for AI workflow experimentation.
- ▶ Deployed a customer retention predictive model using Scikit-learn, integrating it into production workflows.
- ▶ Contributed to model reliability testing, data validation, and iterative AI improvement processes.

Frontend Developer (Intern) · **Marvels Creative Technology***Dec 2024 – Apr 2025*

- ▶ Contributed new features to a Job Portal using Next.js and Tailwind CSS, driving a 15% increase in user engagement.

Technical Content Specialist · **GAOTEK Inc.***Jul 2024 – Dec 2024*

- ▶ Produced technical WordPress content for hardware and software products, improving product discoverability online.

AI & TECHNICAL EVALUATION EXPERIENCE

- ▶ Actively building AI-powered applications using FastAPI, LangChain, LangGraph, RAG pipelines, and LLM APIs (OpenAI, Anthropic) — including autonomous agent workflows and multi-step reasoning systems.
- ▶ Experienced in evaluating AI-generated code outputs for correctness, logical consistency, readability, and performance implications across JavaScript/TypeScript and Python codebases.
- ▶ Skilled in prompt engineering — designing, testing, and iteratively refining prompts to improve LLM output quality and reliability.
- ▶ Familiar with RLHF concepts, AI response ranking, hallucination detection, and reasoning validation workflows.
- ▶ Strong ability to explain complex technical concepts through step-by-step reasoning, debugging walkthroughs, and implementation breakdowns.
- ▶ Proficient in identifying edge cases, async issues, type inconsistencies, and architectural flaws in JavaScript/TypeScript applications — directly applicable to AI code review and training data workflows.